

Food Delivery Eco-System: When Platforms Get Enterprises and Gig-Workers to Implicitly Cooperate

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ABSTRACT

Abstract: The gig-economy relies on intermediation platforms that connect customers with gig-workers. The food delivery eco-system includes four actors: platform, restaurant, courier and customer. It differs from other well-studied “gig” platforms such as Uber or Amazon Mechanical Turk as it brings together a business and a gig-worker. Based on our qualitative study involving couriers and restaurants, we demonstrate that a cooperation is needed between restaurants and couriers to perform food delivery. The way platforms are designed does not support this cooperation as it is affected by the waiting times at the restaurants and the rating system. A dedicated space where both couriers and restaurants could directly give feedback and information could enhance their understanding of the situation and thus enable cooperation.

CCS CONCEPTS

• : Human-centered computing; Collaborative and social computing;

KEYWORDS

Online food delivery, Gig-economy, Cooperation, Restaurant, Courier

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1 INTRODUCTION

The gig-economy relies on intermediation platforms that connect requesters or customers with “gig-workers” who could be drivers, crowd-workers or couriers [10]. So far, the most studied platforms, such as Amazon Mechanical Turk (AMT), Upwork, Uber or Lyft

are composed of three major actors. For the crowd-work platforms (AMT, Upwork), the actors are the requesters (companies/entities that offer micro-tasks to be done), self-employed crowd-workers and the platform [8]. For ride hailing, the three major actors are: customers requesting a trip, self-employed drivers (using their car) and the platform [3]. However, in the case of food delivery, it involves four actors: customers (ordering food), restaurants (preparing the dishes), self-employed couriers (with their bike, motorcycle or car) and the platform. This new configuration implies a cooperation between a business, i.e. the restaurant and the courier, towards a shared objective of satisfactorily delivering the order to the customer [5]. In this contribution, we explore how global food delivery platforms such as Uber Eats, Deliveroo or Just Eat support this type of cooperation of couriers with restaurants and what are the side effects.

The prior knowledge on the relationship between gig-workers and platforms can be applied to the relationship between food delivery platforms and self-employed couriers. Here we refer to the way couriers are incentivized to work for some time and place, the lack of transparency on the consequences of rating and the assignment of tasks [4]. Similarly, the self-employed status of couriers puts them in a precarious situation [7]. In the delivery eco-system, the couriers must also include the constraint of other workers (the restaurants) in their activity.

2 METHOD

In order to understand the functioning of the eco-system, we interviewed 16 couriers, 7 restaurant managers and did observations of the couriers (2 shifts) as well as the restaurants (5 shifts). Couriers were recruited through an online survey shared on social networks and by approaching them directly on the field. Restaurants in our (mid-sized French) city were invited and volunteered to participate in the study. Couriers and restaurants worked either with global platforms (such as Uber Eats, Deliveroo, Just Eat, Stuart etc.) or smaller local platforms (enterprises or co-operatives), as indicated in Table 1. All the interviews were transcribed and analyzed from an ethnomethodological perspective. We had no theoretical orientation and were interested in how participants expressed their usage, activities, understandings and opinions about the food delivery service in their own words [9]. The interviews lasted between 12min to 98min, with an average of 55 minutes. A significant observation that stood out was the reported frictions between the couriers and the restaurants showing the difference in the objectives of these actors and how the global platforms poorly support their cooperation.

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Table 1: Details of participants involved in the study. For the couriers and restaurants, we indicate the type (global or local) and the number of platforms they have worked with.

Couriers	Years/Months of experience on the job	No. of global platforms	No. of local platforms	Restaurants	No. of global platforms	No. of local platforms
Courier 1	3 – 6 months	1	0	Restaurant 1	3	0
Courier 2	More than 1 year	2	1	Restaurant 2	3	1
Courier 3	3 – 6 months	1	0	Restaurant 3	1	0
Courier 4	3 – 6 months	1	0	Restaurant 4	1	1
Courier 5	6 – 12 months	3	0	Restaurant 5	1	1
Courier 6	More than 1 year	1	1	Restaurant 6	3	1
Courier 7	More than 1 year	1	0	Restaurant 7	0	1
Courier 8	3 – 6 months	2	0			
Courier 9	More than 1 year	2	2			
Courier 10	More than 1 year	2	1			
Courier 11	More than 1 year	1	1			
Courier 12	More than 1 year	1	1			
Courier 13	More than 1 year	2	2			
Courier 14	More than 1 year	1	1			
Courier 15	More than 1 year	2	1			
Courier 16	More than 1 year	2	1			

3 FINDINGS

In this part, we unpack how cooperation is operating between the couriers and the restaurants. The cooperation is affected by elements induced by the platform's design or the status of couriers. Without having any formalized contracts, there still exists an interdependency between couriers and restaurants. The restaurants subscribe to a delivery service proposed by the platforms that rely on independent workers, the couriers. On the other side, couriers are self-employed partners of the platform. Platforms claim they can work whenever they want, just by opening the app and are paid an amount of money for each delivery they perform. This means that once online, the couriers want to deliver as many orders as possible to make money, introducing high pressure on their time.

3.1 The trouble with estimating waiting times

When the couriers have accepted to pick up an order, they ride to the restaurant and, in most cases, have to wait for it. For instance, Courier 5 explains, *"And then either the order is ready, and you pick it up, but in most cases, you have to wait and that is the worst thing that happens for a courier, to have to wait each time to pick up the orders"*. To avoid waiting times, couriers adopt different strategies. Four interviewees mention that they refuse notifications of order depending on the restaurant or they cancel it if the delay is too long in front of the restaurant. *"I avoid restaurants where there are a lot of people and waiting"* (Courier 2). *"Some restaurants we know well for being late, such as (this specific fast-food chain). If we realize that it is going to be very long, we can cancel the order to go and take another order at another restaurant"* (Courier 7). However, rejecting orders is at their own risk because they might well wait some (unknown) time before receiving another order or even end up waiting again for the restaurant on the next order they will accept to deliver. *"There are a lot of couriers who spend most of their time waiting for*

orders" (Courier 11). They can even take the opportunity in the waiting time to manage another order on another platform. *"If I accept (this fast-food chain) order, I know that on another platform, I can deliver another order at the same time. (So) I will not wait for 15 minutes. I'll deliver another order while waiting. Then I'll come back, and if I wait for 3-4 minutes, it is not a big deal"* (Courier 5).

The manager of the restaurants also perceives this time pressure experienced by the couriers. As said by the manager of Restaurant 2, *"when I really see that it is a problem, I will tell them, well, it is better that you cancel because then you will have to wait. Some (couriers) decide to wait and come to you every two minutes. Is it ready? Is it ready?"*. According to several couriers, some restaurant managers misestimate the waiting time. *"sometimes they tell you 2min when it is 10min. Sometimes 5min, for them it's the standard, 5min means from 0 to 20min"* (Courier 8). According to her, it is because *"they want to keep the couriers at the restaurant door"*. From the restaurant managers' perspective, some try to provide the waiting time even if it is difficult to be precise, especially during the rush hours. Others would always give the same waiting time: *"in general, we give them a big 8 to 10 minutes"* (Restaurant 7). This phenomenon is sometimes worsened when the platform sends order notification almost simultaneously to the restaurant and the courier. *"Sometimes they (the couriers) have the order almost at the same time as we have it, so he (the courier) has the impression that we are going to go and do it right away except that in front of him there are 45 (orders)"* (Restaurant 2). This situation directly results from the incentivized work model of the couriers. It could be improved by payment per hour rather than per task as mentioned by few couriers working on local platforms. *"For example, when we go to the restaurant we will go from 7 pm to 9 pm ensured with an hourly rate of 15€ per hour sure. Even if there is 0 order, you are sure to make 30€"* (Courier 6).

3.2 Rating: a way to ensure smooth couriers-restaurants cooperation?

On the platform, couriers and restaurants can rate each other. “We (couriers) can rate restaurants and customers” (Courier 4). Couriers can rate restaurants, and four participants explain they have used this feature. “Yes, I have done it many times in restaurants, especially [...] Normally, in principle, the order should be ready when you receive it (the notification) on the phone but this is not the case at all. You have to wait 15-20 minutes in front of the restaurant” (Courier 8). Conversely, the restaurants can rate couriers: “it includes the pick-up of the order, the way you stand in the restaurant, if you are not too noisy, cumbersome” (Courier 2). In extreme cases, the manager of Restaurant 1 explains that they could act beyond rating: “in the worst case, I can even call a dispute department so that (this global platform) takes sanctions on their side, [...] even if I think that it is almost never done”. This quote shows the limited perceived impact on such ratings and the extra work it implies for the restaurant to provide feedback. The interviews highlighted that neither the couriers nor the restaurants are convinced that using the rating system is an efficient way to change bad behaviors and ensure a smooth cooperation. There is only one exception; a courier explains they collectively tried to impact one restaurant’s waiting time. “Putting together a bunch of people (couriers) and putting up red thumbs, it makes (this global platform) wake up, call them (the restaurant), and there are little things that have come in improvement” (Courier 7). We observed that the rating proposed by the platforms is perceived to be ineffective in supporting satisfying cooperation between the couriers and the restaurants.

3.3 Poor communication affecting the global service quality

From our observations of the food pick-ups at the restaurants, there is little communication between the courier and the restaurant. The most common pattern is to see the courier entering the restaurant, pronouncing a name or a number and taking a bag if it is ready. “We arrive, we report (our presence), when the bag is ready they give it to us right away otherwise we wait” (Courier 7). “There must be 10% of the couriers who are polite, they come in, (shout) 800 number, they take the bag they leave. No hello, nothing [...]” (Restaurant 3). This absence of communication beyond what is strictly necessary affects the global quality of service. We develop here two examples where miscommunication induces negative effects for the actors. The first example is the risk of having errors in the orders. Both the couriers and the restaurants agree that it is the restaurant’s responsibility to avoid any error with the food ordered. “We do not open the bags to check what is in them. It is the restaurant owner’s responsibility” (Courier 9). From the interviews and observations, restaurants put a strong effort to avoid errors. In three restaurants, they have dedicated employees to check the content of the bags. “We try to put those (the employees) which are the most experienced one [...] because we have to check (the orders) and we cannot afford mistakes [...] we check everything (the contents of the order)” (Restaurant 2). As the manager of Restaurant 3 reports: “The couriers who come in, see a number, they take it (the bag) because it’s theirs, but it happens that you have two, three or four bags for the same order [...]. (Then) customers call “well I have half (of my order)”, well yes the courier

came in, the bag is gone, but he forgot half the order”. Though the courier is the source of error in this situation, the customer is bound to interpret it as the restaurant’s error. As we observed, the extra effort to check the bags results from the difficulties to fix the errors. “So when the customer is not happy and says I have half of my order, the problem is that I cannot deliver again” (Restaurant 3).

The second example we develop is around the packaging. Some couriers point to difficulties regarding the packaging used by the restaurants. Courier 1 refers to spilled orders because the packaging was not adapted. “Unlike other restaurants, (this fast-food chain) still uses bags and not boxes (for the beverages). So in the bag, it is easy to spill.” To avoid this issue, Courier 1 and Courier 3 refer to some techniques: “I have a towel that I put in my bag if there are drinks or what I’ll be able to wedge things in” (Courier 3). In this situation, the choice of the packaging is the restaurant’s one, but if the order arrives overturned, it is the courier’s responsibility. As an answer to this situation, another participant explains: “It often happens with them (some restaurants), that the orders open, with the drinks, because the cups are perhaps badly closed. I refuse this restaurant for that.” (Courier 2). From the courier’s perspective, he is responsible for delivering the order which is not spilled while depending on the restaurant’s choice of food packaging. The instances of delivering a complete order and providing adequate food packaging suited for delivery are two examples that show the bonded objectives of couriers and restaurants to reach customer satisfaction.

4 DISCUSSION

4.1 Contribution

Given their roles in the food delivery eco-system, couriers and restaurants have to cooperate. The findings show that both parties focus on the efficiency of their tasks but not on their cooperation. The couriers mainly pay attention to the waiting time and the restaurants mostly try to avoid mistakes, while they share a larger objective: customer satisfaction. On the side of the restaurants, this larger objective is clearly claimed and does not change from the one they have concerning service on-site. On the side of couriers, this objective is not clearly stated and seems to result from the possibility for the customer to rate the courier activity. This notion of satisfying the customer for the couriers does not exist as a common goal with the restaurant, rather because of intermediary features imposed by the platform’s design. From the last part of the findings, we understand that couriers’ and restaurants’ activities cannot be bluntly divided into two parts, with the restaurant on one side preparing the food and the courier on the other side delivering the bag. These activities are tied together, and negative actions of one side have consequences on the other side. What is missing so far is a space where both couriers and restaurants could exchange on their respective practices, including the specifics and variability of each day that can have consequences on both side. For instance, one absent employee at the restaurant creates additional delay, or even the notification about a new packaging that the restaurant wants to test (for the sealing). This sharing of information will allow the courier and the restaurant to plan their activities with the knowledge about the constraints of the other party to achieve the larger goal of successful food delivery. Getting inspiration from the prior knowledge in CSCW on *articulation work* [11], we could consider

designing an articulation space to be added to the platforms such as forums [2]. However, the design and implementation of such a feature require further research to understand its effectiveness in supporting cooperation.

4.2 Future research

We plan to extensively understand the activity of *local* platforms, which are relatively small enterprises or co-operatives that have a dedicated team of couriers to deliver food. In many of those organizations, the couriers are paid per hour instead of per task. Their connection with the restaurants also seems quite different from the couriers working for global platforms, as understood from the first insights we collected with couriers and restaurants. We could then design technologies that support local organizations for food delivery, potentially also informing the design of global platforms [1].

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